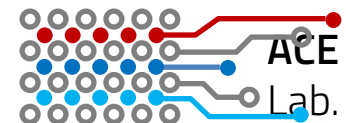


ROS2 응용과정(1장)과 F1논문리뷰

Aaron Yoon

Architecture and Compiler for Embedded Systems Lab.
School of Electronics Engineering, KNU, KOREA

2026.07.07



ToC

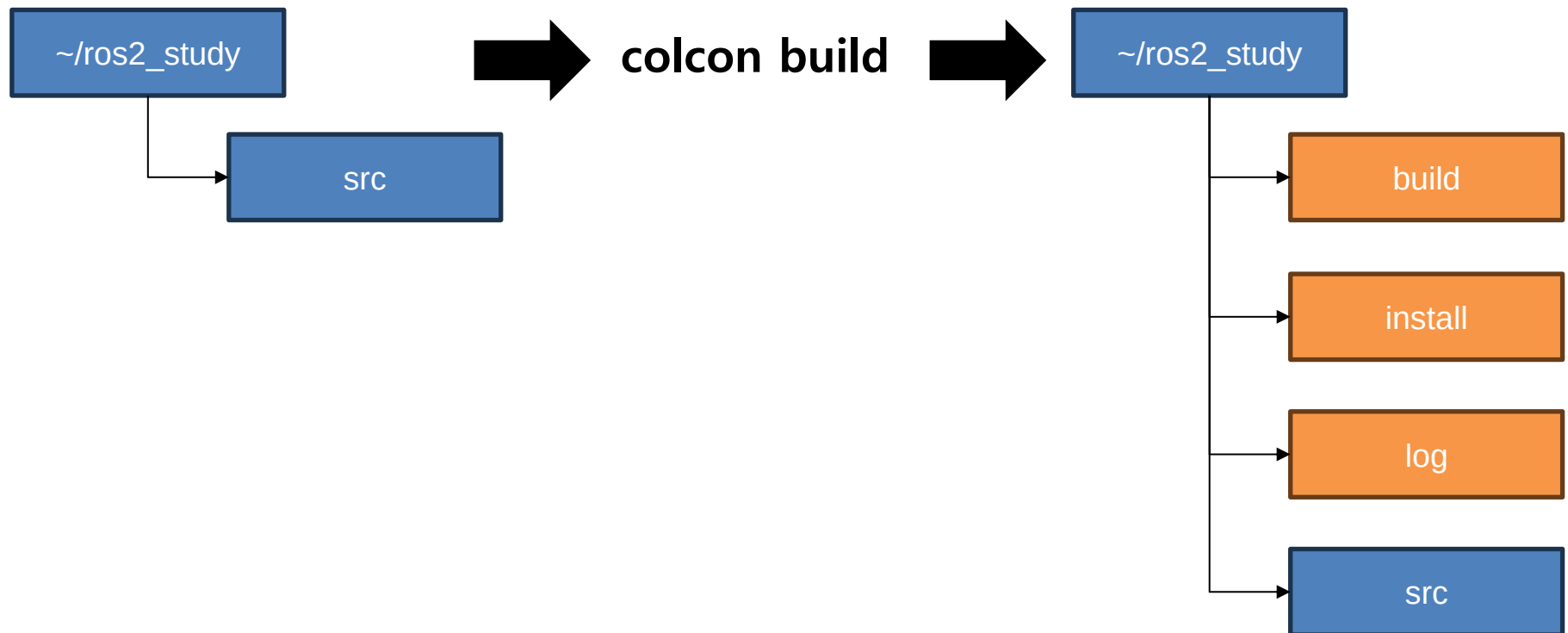
1. Package, Node 만들어보고 Node 실행하기
2. 커스텀 package에서 topic 구독해보기
3. 커스텀 package에서 topic 발행 기능 넣어보기

1.Package,Node 만들어보고 Node 실행하기

```
mkdir -p ~/ros2_study/src
```

```
cd ~/ros2_study
```

```
colcon build
```



1.Package,Node 만들어보고 Node 실행하기

```
cd ~/ros2_study/src
```

```
ros2 pkg create ₩
```

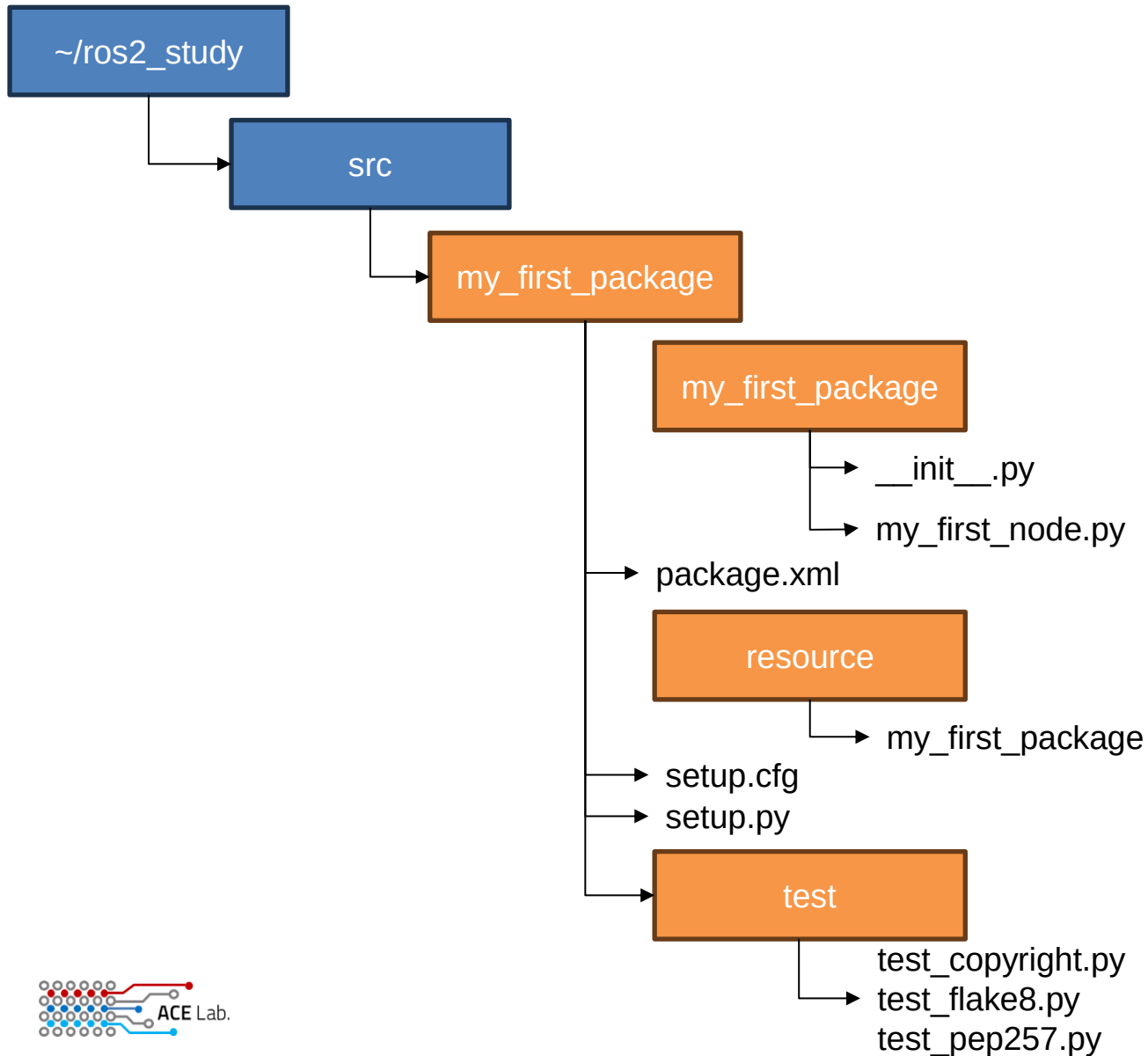
```
--build-type ament_python ₩
```

```
--node-name my_first_node ₩
```

```
my_first_package
```

```
colcon build
```

1. Package, Node 만들어보고 Node 실행하기



2.커스텀 package에서 topic 구독해보기

Lorem ipsum

3.커스텀 pacakge에서 topic 발행 기능 넣어보기

Lorem ipsum

4.F1 논문 리뷰

F1/10: An Open-Source Autonomous Cyber-Physical Platform

Matthew O'Kelly*, Houssam Abbas**, Jack Harkins, Chris Kao, Yash Vardhan Pant & Rahul Mangharam

Department of Electrical and Systems Engineering, University of Pennsylvania, USA
{mokelly, harkj, chriskao, yashpant, rahulm}@seas.upenn.edu

** School of Electrical Engineering and Computer Science, Oregon State University, USA
houssam.abbas@oregonstate.edu

Varundev Suresh Babu*, Dipshil Agarwal & Madhur Behl

Department of Computer Science, University of Virginia, USA
{varundev, dpa4va, madhur.behl}@virginia.edu

Paolo Burgio & Marko Bertogna

Dept. of Physics, Informatics & Mathematics, University of Modena & Reggio Emilia, Italy
{marko.bertogna, paolo.burgio}@unimore.it

* These authors contributed equally

ABSTRACT

In 2005 DARPA labeled the realization of viable autonomous vehicles (AVs) a *grand challenge*; a short time later the idea became a *moonshot* that could change the automotive industry. Today, the question of *safety* stands between *reality* and *solved*. Given the right platform the CPS community is poised to offer unique insights. However, testing the limits of safety and performance on real vehicles is costly and hazardous. The use of such vehicles is also outside the reach of most researchers and students. In this paper, we present F1/10: an open-source, affordable, and high-performance 1/10 scale autonomous vehicle testbed. The F1/10 testbed carries a full suite of sensors, perception, planning, control, and networking software stacks that are similar to full scale solutions. We demonstrate key examples of the research enabled by the F1/10 testbed, and how the platform can be used to augment research and education in autonomous systems, making autonomy more accessible.

1 INTRODUCTION

Progress in cyber-physical systems (CPS) requires the availability of robust platforms on which researchers can conduct real-world experiments and testing. Unfortunately, a vast majority of CPS experiments are done in isolation - either completely in simulation,

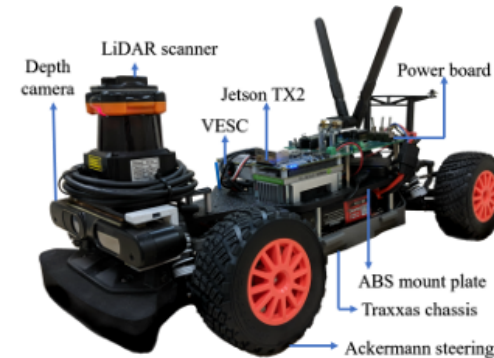
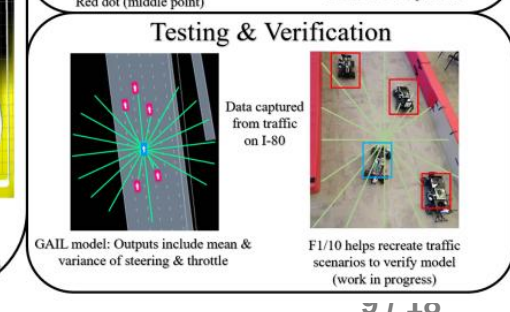
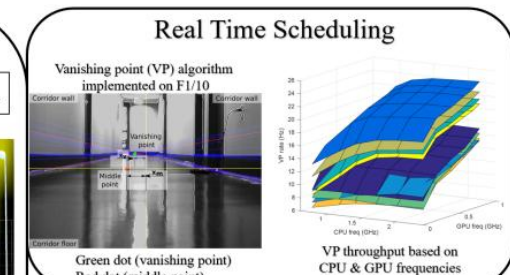
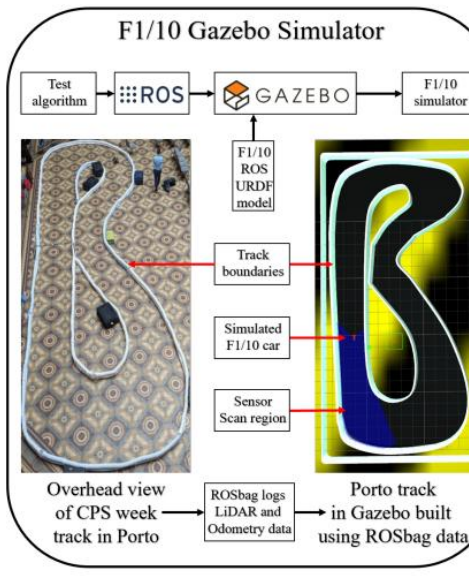
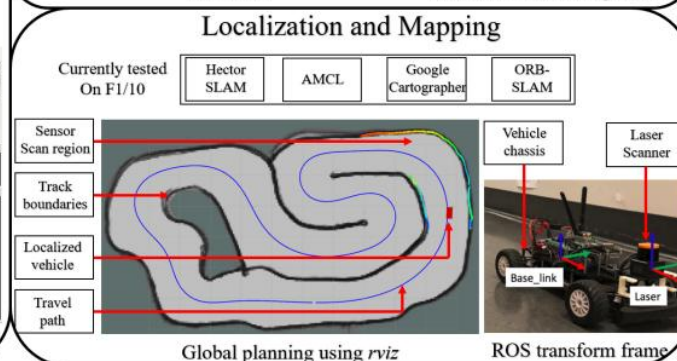
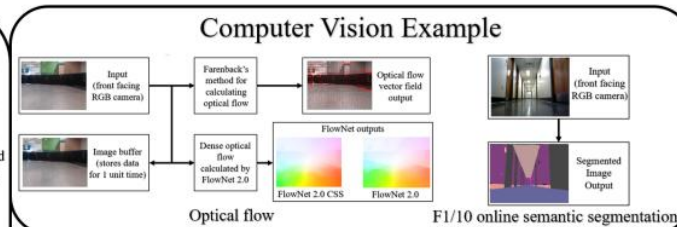
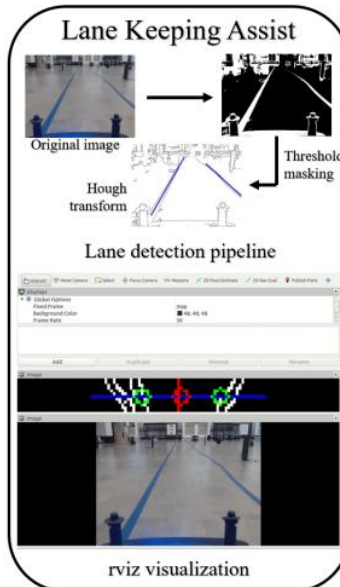
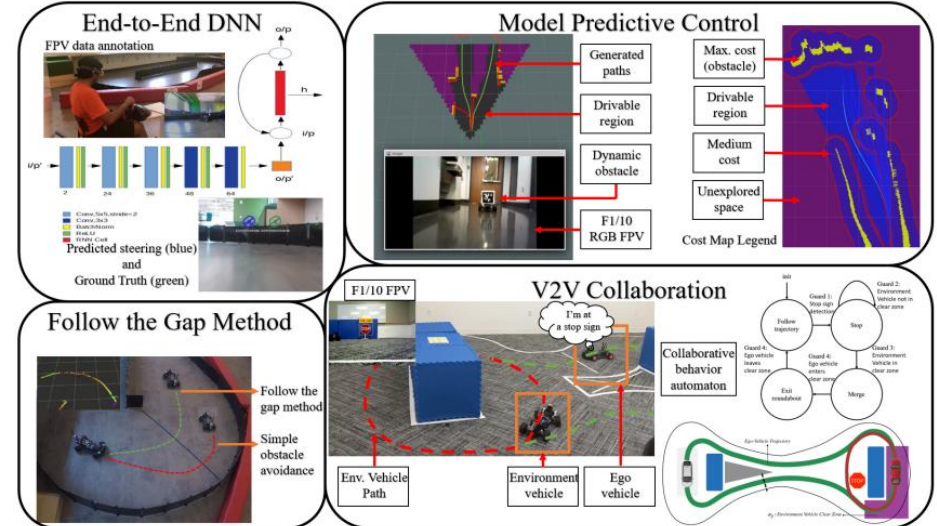
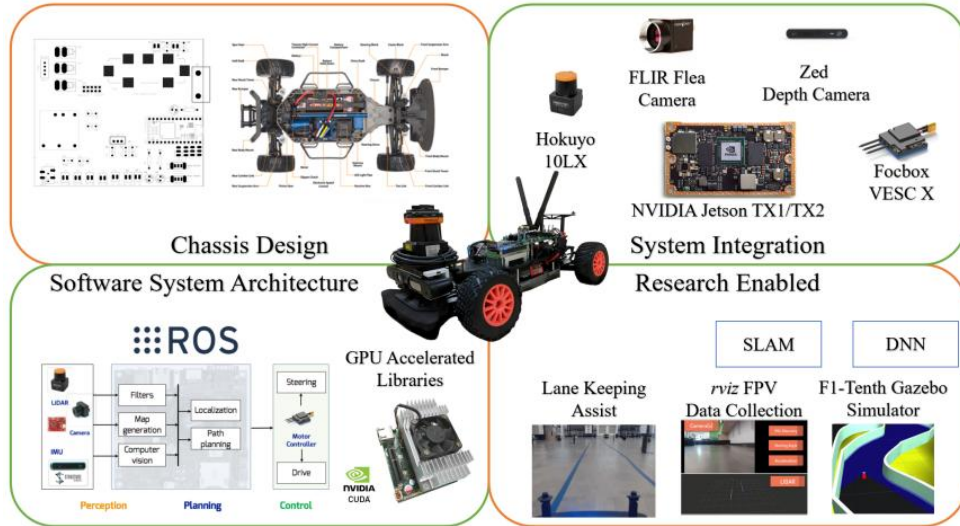


Figure 1: It takes only a couple of hours fully to assemble a F1/10 autonomous racecar, using detailed instructions available at <http://f1tenth.org/>

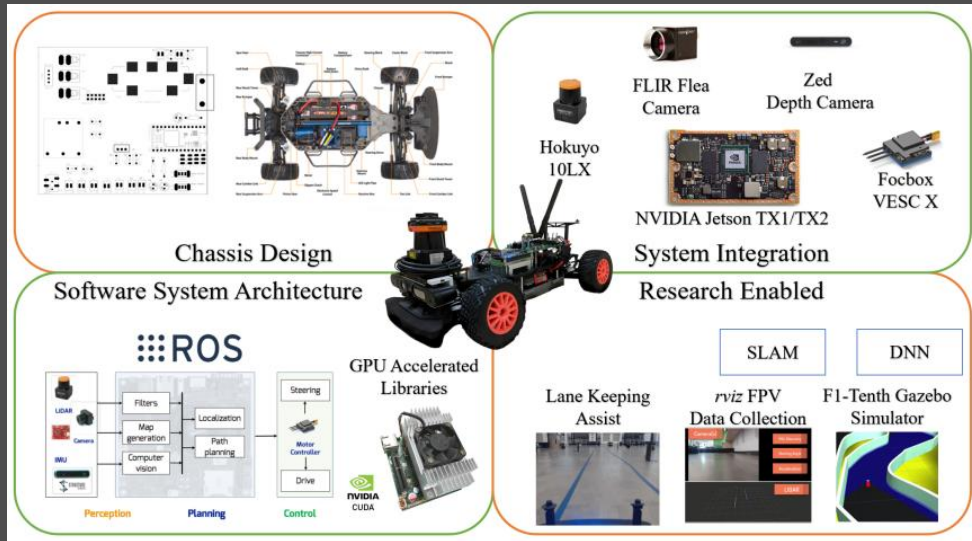
vehicles that will alter our roads and lay the groundwork for future smart cities.

Today, conducting research in autonomous systems and AVs requires building one's own automotive testbed from scratch. Sometimes researchers must enter into restrictive agreements with auto-

4.F1 논문 리뷰



4.F1 논문 리뷰



1. 차량 두뇌: NVIDIA Jetson TX2

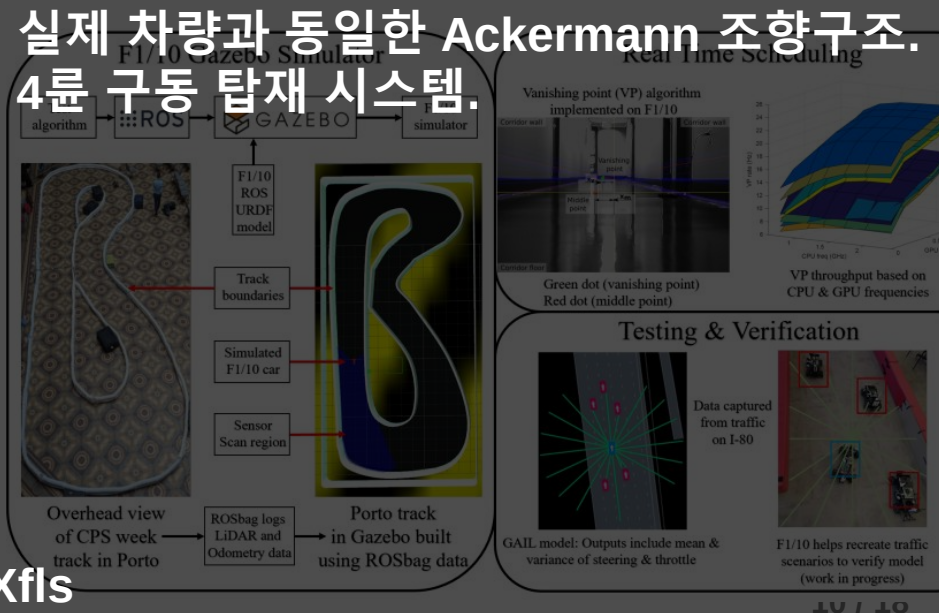
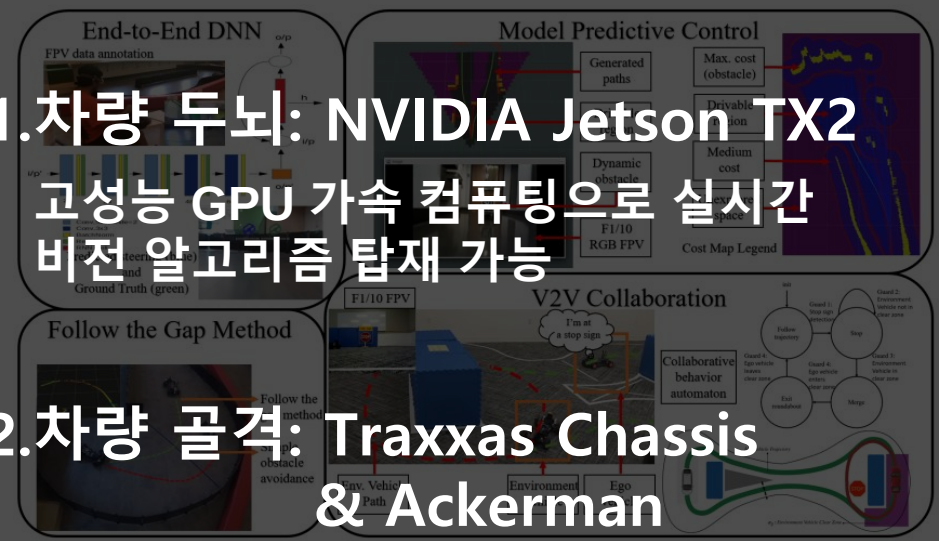
고성능 GPU 가속 컴퓨팅으로 실시간 비전 알고리즘 탑재 가능

2. 차량 골격: Traxxas Chassis & Ackerman

실제 차량과 동일한 Ackermann 조향구조. 4륜 구동 탑재 시스템.

3. 차량 감각: Hokuyo 10LX & ZED Camera(3D Depth Camera)

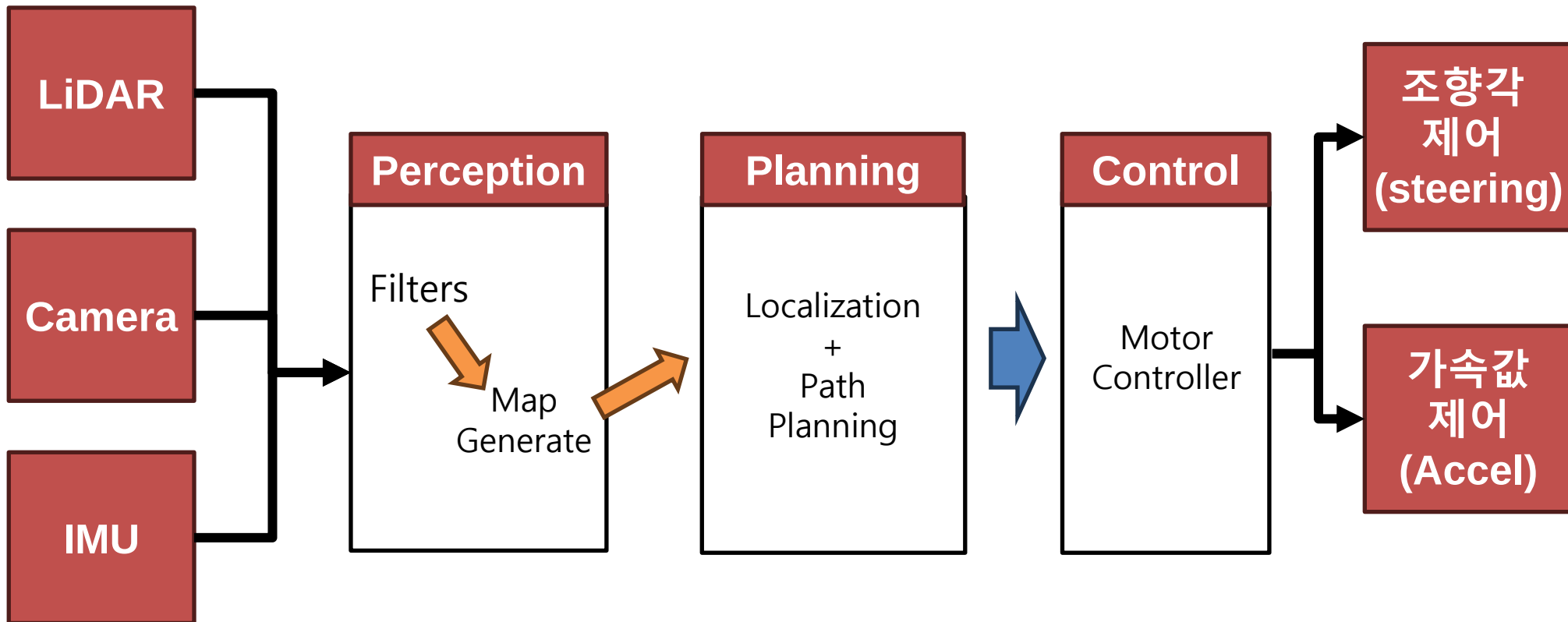
실제 차량과 동일한 Ackermann 조향구조. 4륜 구동 탑재 시스템.



https://www.youtube.com/watch?v=OD_dflZXfls

4.F1 논문 리뷰

SW Architecture: 모듈화된 ROS Runtime Environment

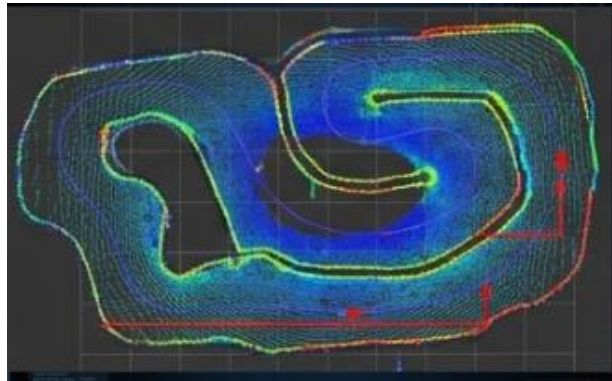


※ ROS 기반의 철저한 모듈화 설계.
엔지니어 입장에서, 전체 시스템을 새로 짤 필요없이, 연구하고자 하는
단일 모듈만 교체하여 즉시 테스트 가능

4.F1 논문 리뷰

연구 활용 (1) : 실시간 인지 및 매핑(Perception)

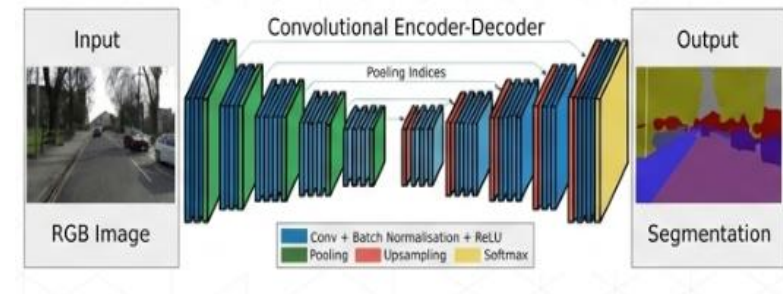
3D SLAM & Mapping



차선 유지
(Lane Keep Assist)



컴퓨터 비전(CV)



RGB Image(Input)
=> E2E => Segmentation(Output)

4.F1 논문 리뷰

연구 활용 (2) : 계획 및 제어 알고리즘 검증

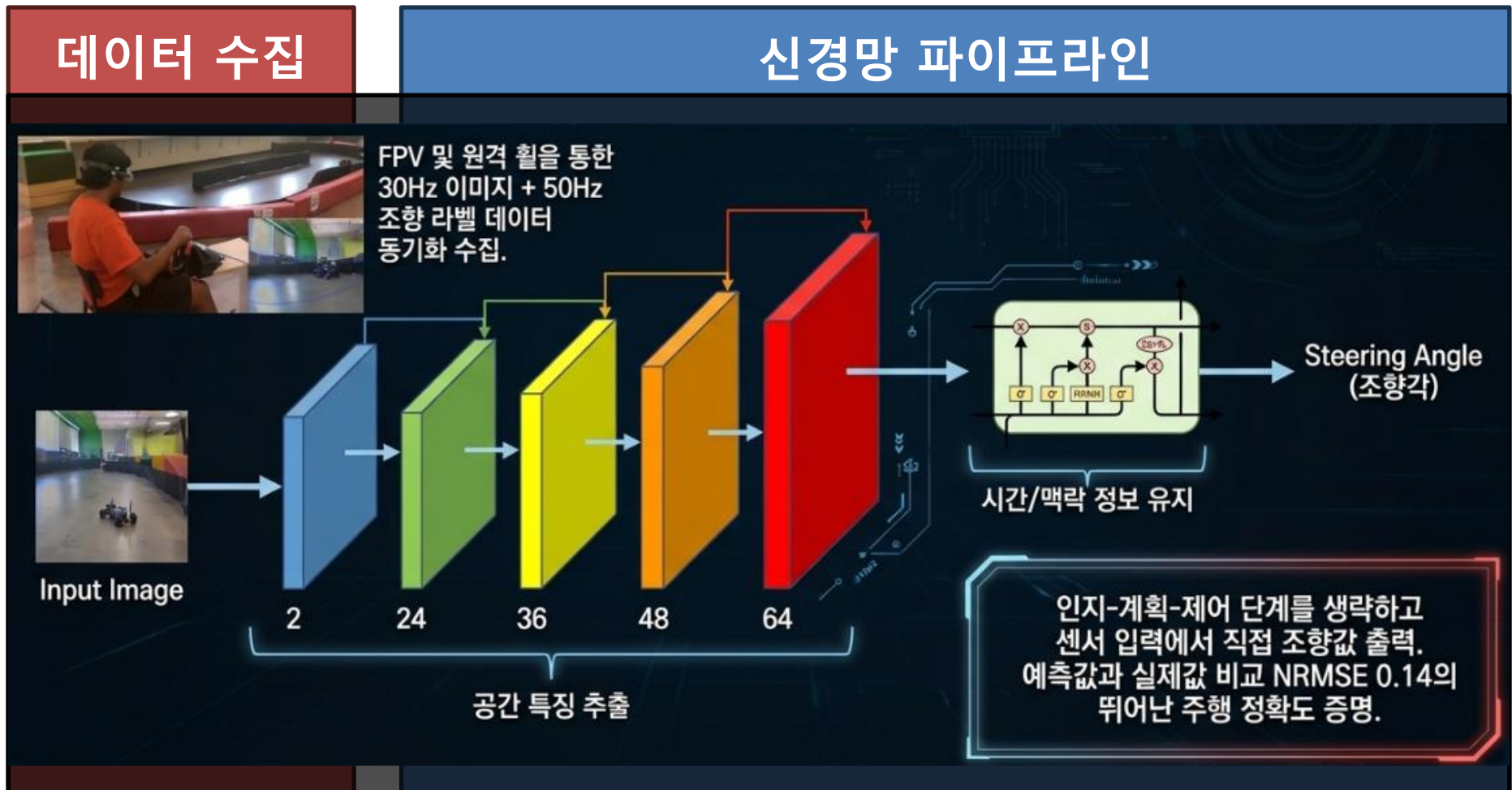
입력
이미지

F1신경망의 구조

CNN+LSTM(장단기 메모리 신경망)을 결합한 하이브리드 구조!

4.F1 논문 리뷰

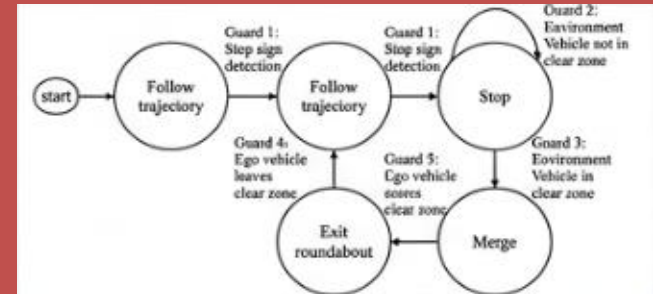
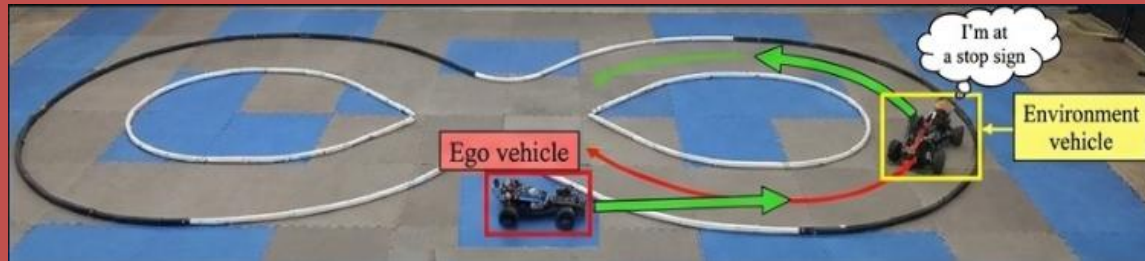
연구 활용 (3) : 종단간 딥러닝(End-to-End DNN)



4.F1 논문 리뷰

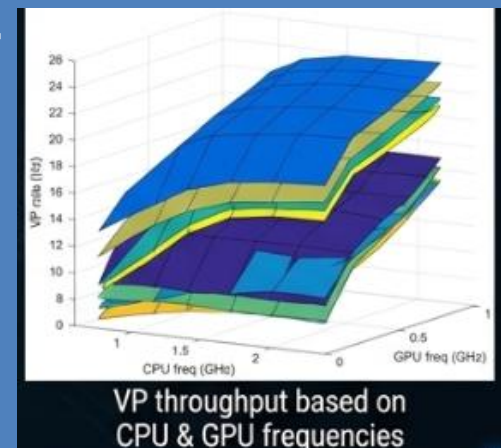
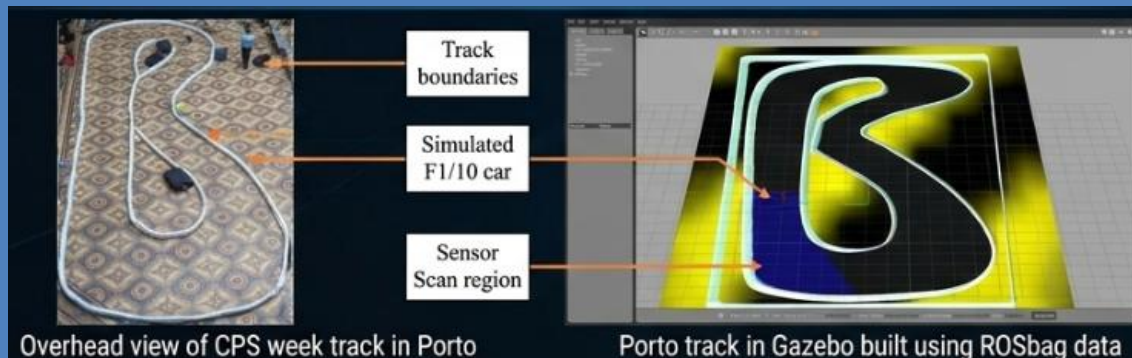
연구 활용 (4) : V2V 통신 및 시뮬레이션 확장

차량 간 협력 주행 (V2V Communication)



사각지대 교차로 시나리오) TCP Client/Server Node를 통해 차량 간 객체 리스트를 실시간으로 공유, 충돌을 방지하는 협력 거동 오토마톤 설계

ROS Gazebo 시뮬레이션 및 시스템 자원 최적화



현실 데이터 기반 사전 테스트 환경 구축, CPU/GPU 간의 실시간 스케줄링을 통한 연산 효율 최적화를 증명.

향후계획

Lorem ipsum

Lorem ipsum

Lorem ipsum

Lorem ipsum

Lorem ipsum

공간지능의 패러다임 전환

2020년을 기점으로 공간지능의 패러다임은 아래와 같이 크게 변화함.

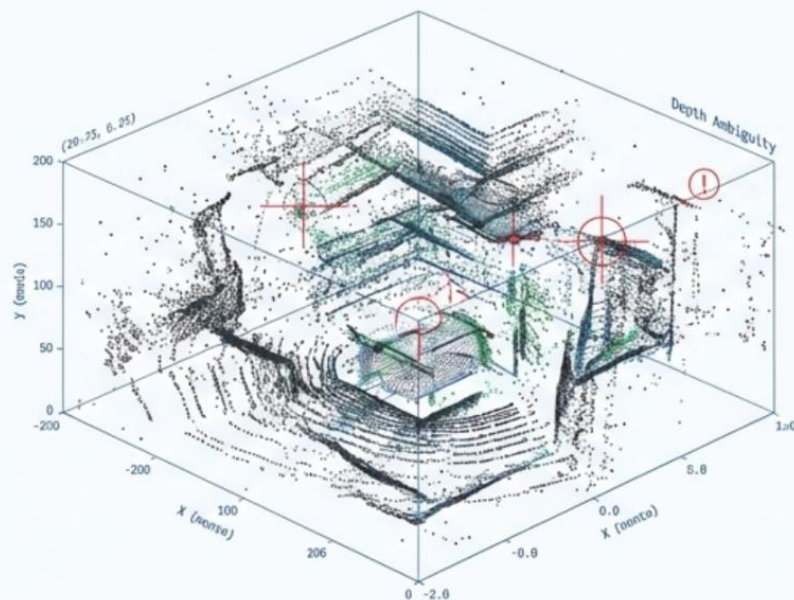
기하학 SfM/MVS



E2E AI

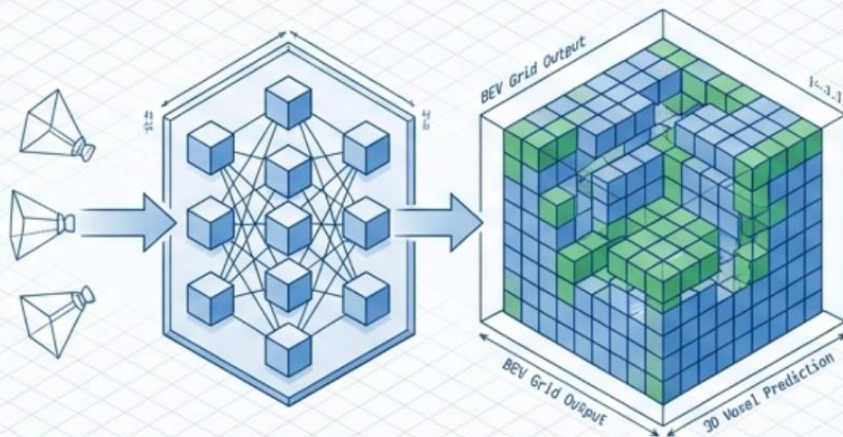
기하학적 복원 방식 (2000년대 후반)

- 핵심 기술: SFM, MVS
- 치명적 한계: 깊이 정합 모호성(Depth Ambiguity)으로 인한 성능 저하
- 연산 속도: 수식적 계산 병목으로 실시간 처리 제약



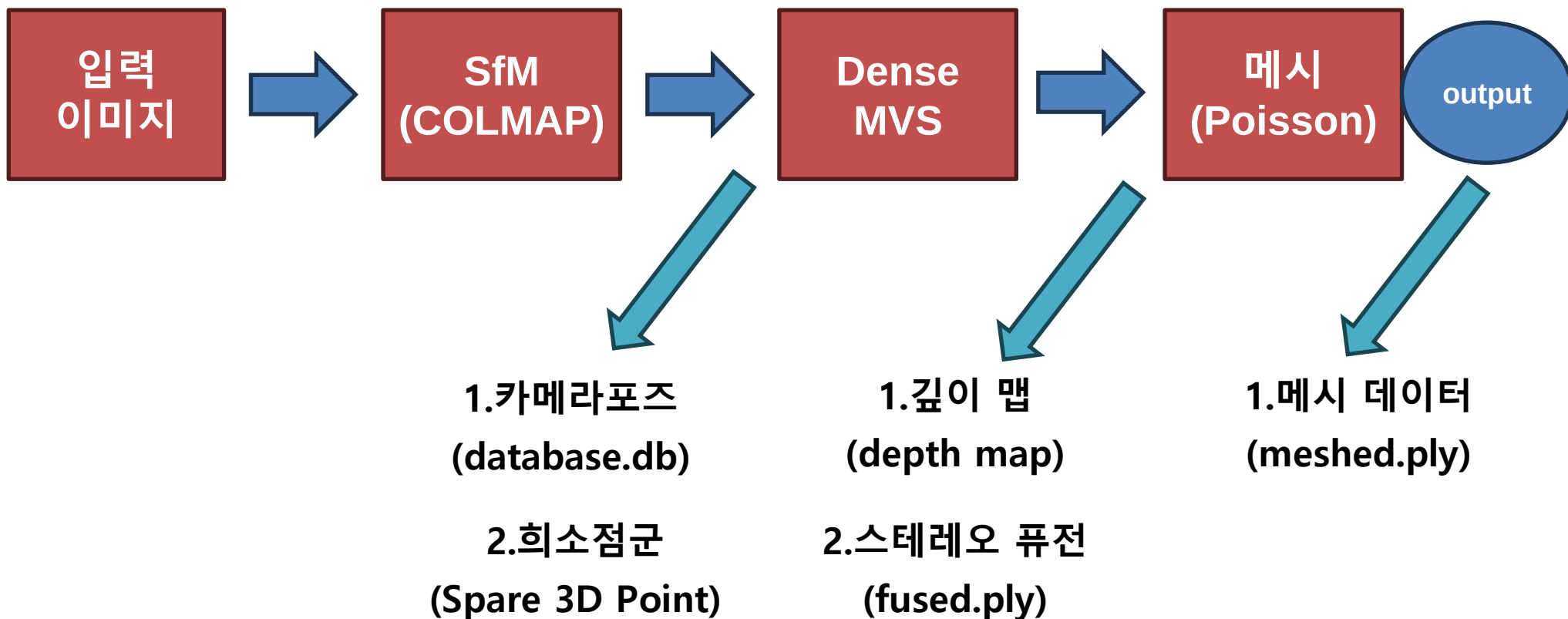
AI / CNN 기반 복원 (현재)

- 핵심 기술: 엔드투엔드(End-to-End) 신경망 학습
- 돌파구: 깊이 및 정합 모호성을 우회적으로 해결
- 성능 도약: 다중 카메라 영상에서 BEV 격자 및 3D 복셀 직접 예측



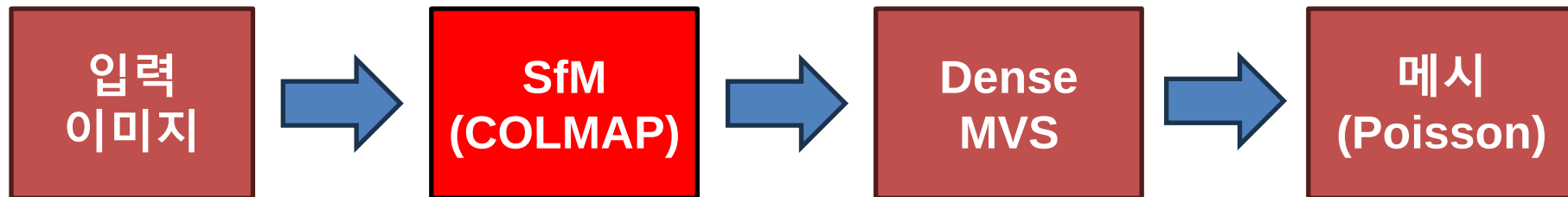
이미지 기반 3D 재구성

전통적 기법: (1)SfM => (2)MVS



이미지 기반 3D 재구성

전통적 기법: (1)SfM => (2)MVS



1.SfM(Structure from Motion) – 희소(sparse) 복원

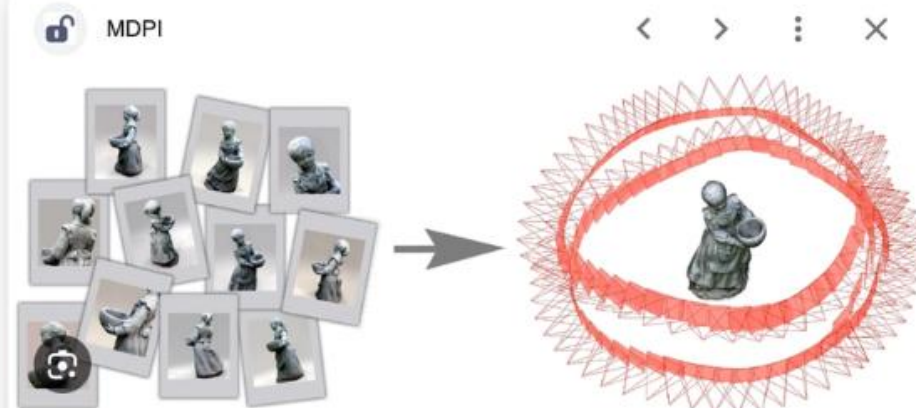
Structure from Motion

SIFT Descriptor로 특징 검출/매칭

카메라 내/외부 파라미터 추정

희소 3D 점 생성

Bundle Adjustment



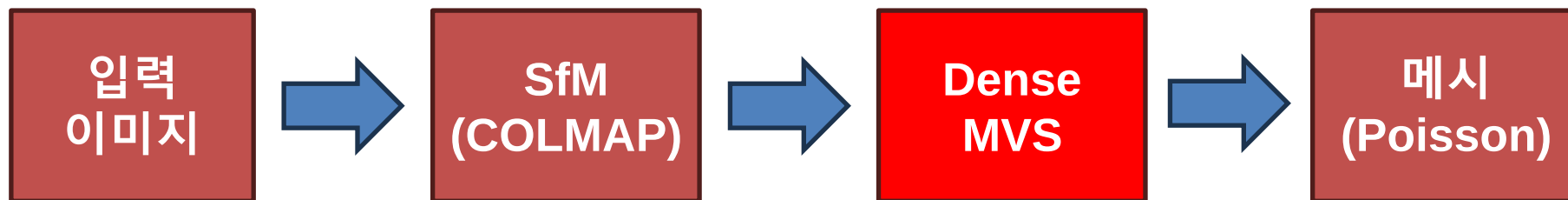
Evaluating the Performance of Structure from Motion Pipelines

저작권 보호를 받는 이미지일 수 있습니다. 자세히 알아보기

방문 >

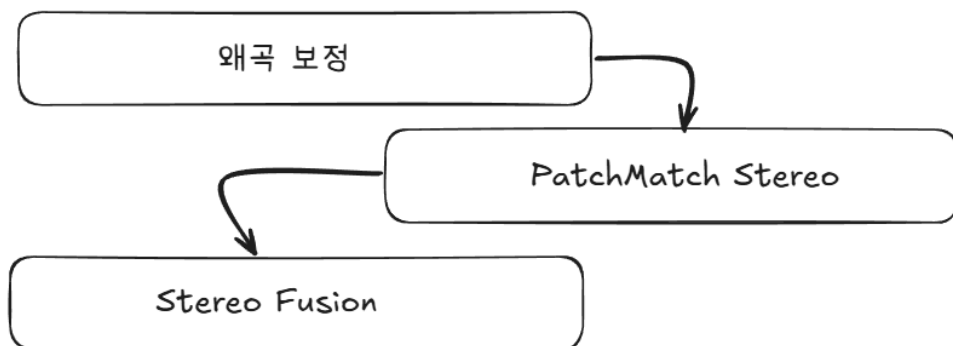
이미지 기반 3D 재구성

전통적 기법: (1)SfM => (2)MVS

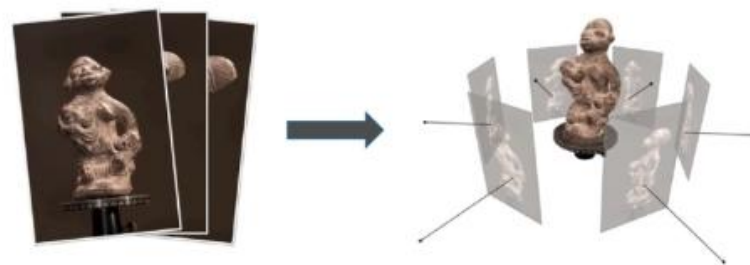


2.MVS(Multi-View-Stereo)

Multi-View-Stereo



Unsupervised MVS reconstruct 3D Object from a set of images without any GT



RC-MVSNet: Unsupervised Multi-View Stereo with Neural Rendering
Di Chang et al. ECCV2022

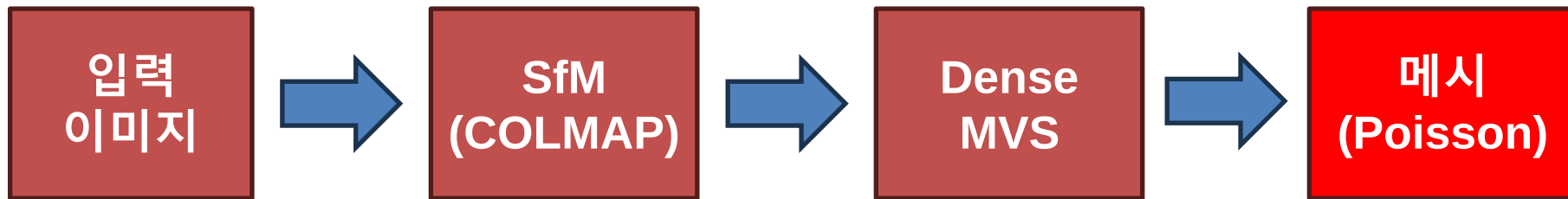
TUM VISUAL COMPUTING

RC-MVSNet: Unsupervised Multi-View Stereo with Neural Rendering (ECCV'22)

[시청하기 >](#)

이미지 기반 3D 재구성

전통적 기법: (1)SfM => (2)MVS



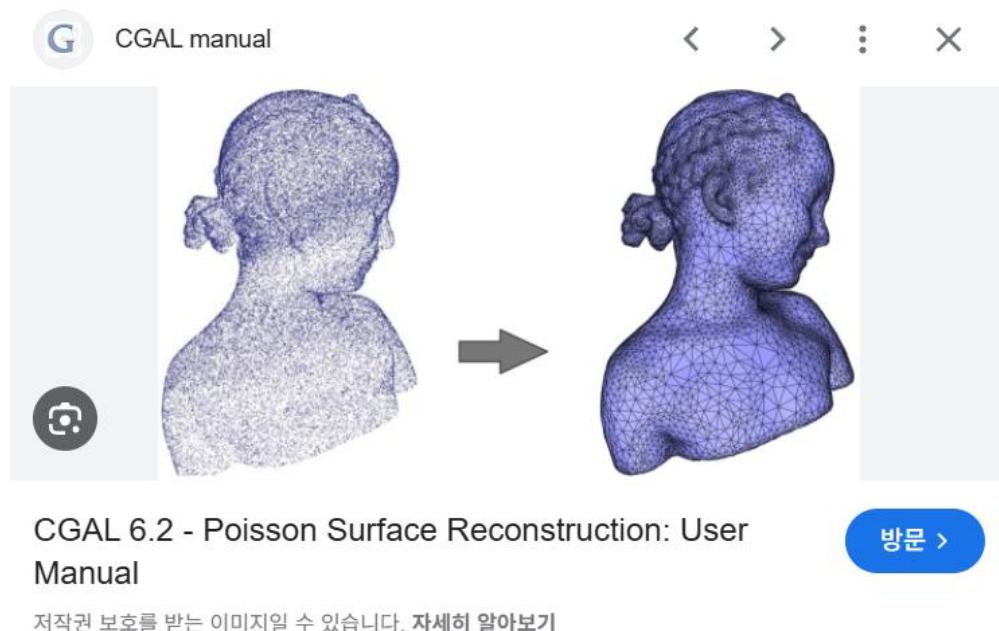
3.Poisson 메시 재구성

Poisson Mesh Reconstruction

조밀 점군의 위치 + 법선 입력

연속 표면 복원

Screened Poisson으로 빈 영역 매꾸



현대판, 이미지 기반 3D 재구성

전통적 3D 복원의 한계: 깊이 정합 모호성!! -> AI를 통해 이 문제 해결!

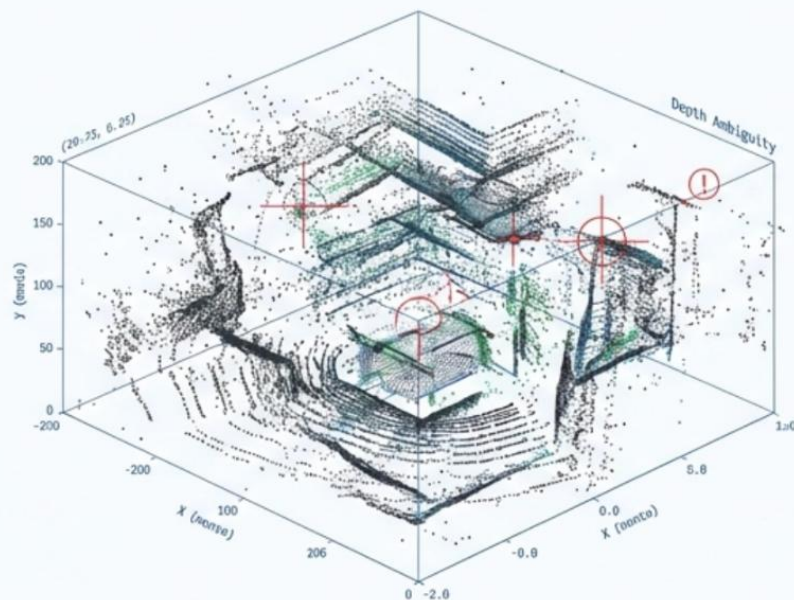
기하학 SfM/MVS



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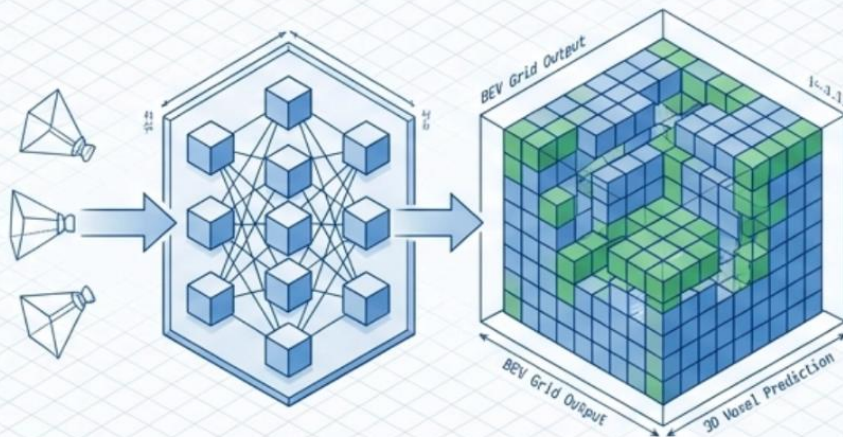
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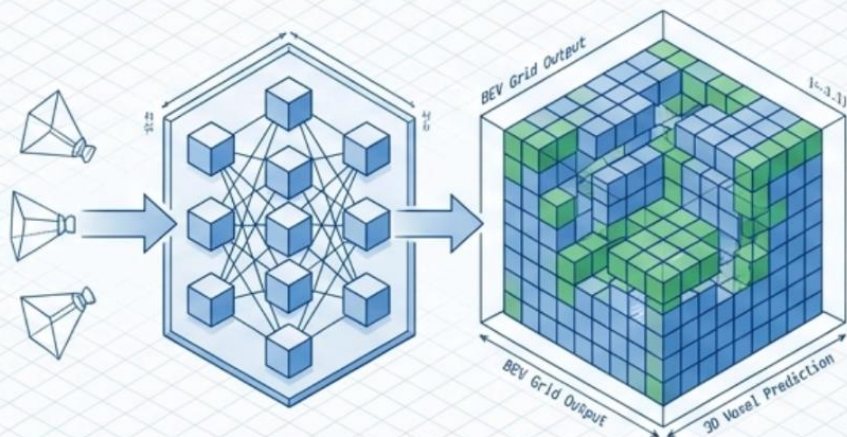


현대판, 이미지 기반 3D 재구성

2D -> BEV 변환

AI / CNN 기반 복원 (현재)

- 핵심 기술: 엔드투엔드(End-to-End) 신경망 학습
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=> 앞으로 AI를 활용한 이미지 기반 3D 재구성은
“View-Transformer”라고 부름.

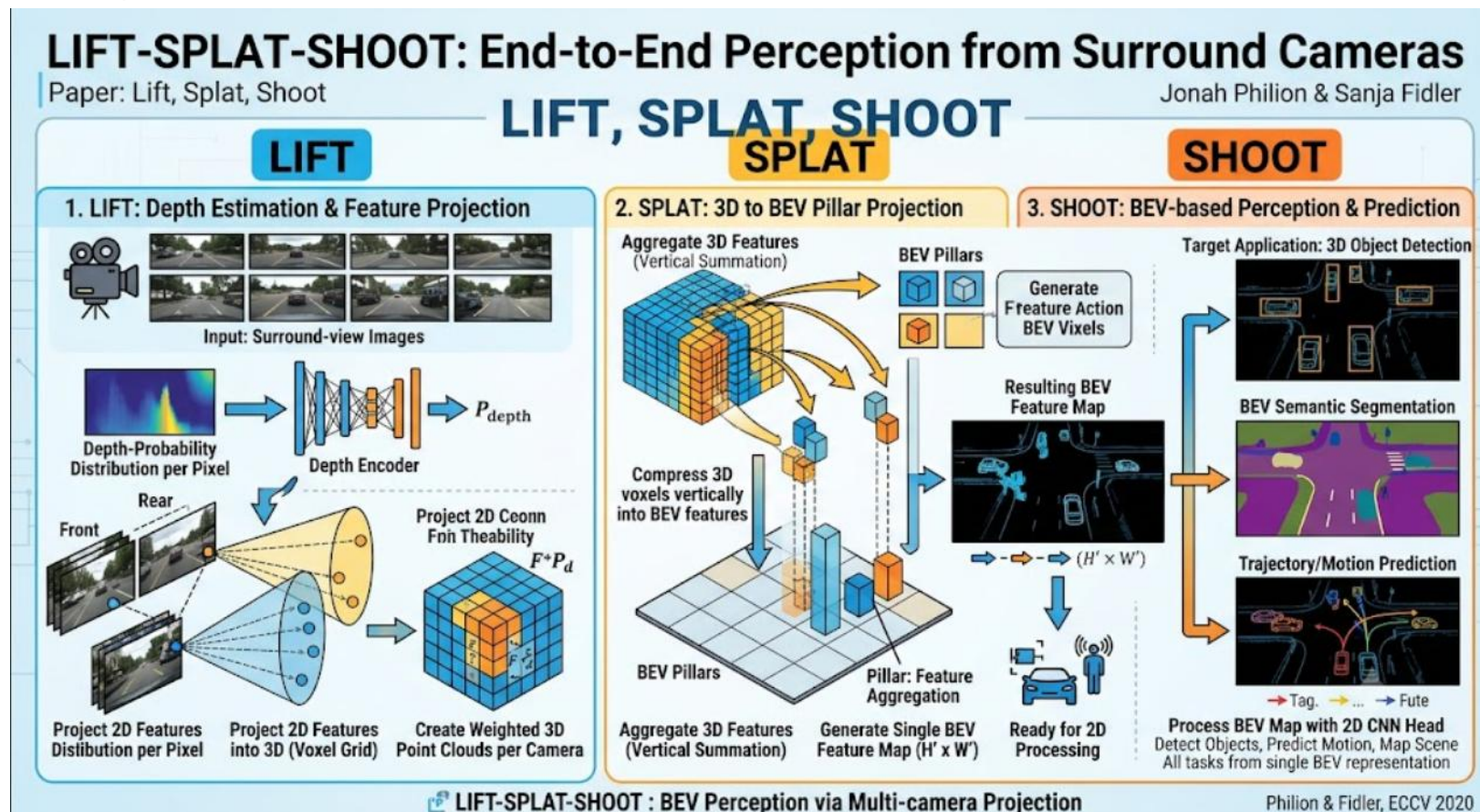
View-Transformer에는 두가지 분류가 있음.

분류(1): 순방향 투영(Forward Projection)
-> LSS(Lift-Splat-Shoot)

분류(2): 역방향 투영(Backward Projection)
-> BEV Former

순방향투영방식: LSS(Lift-Splat-Shoot)

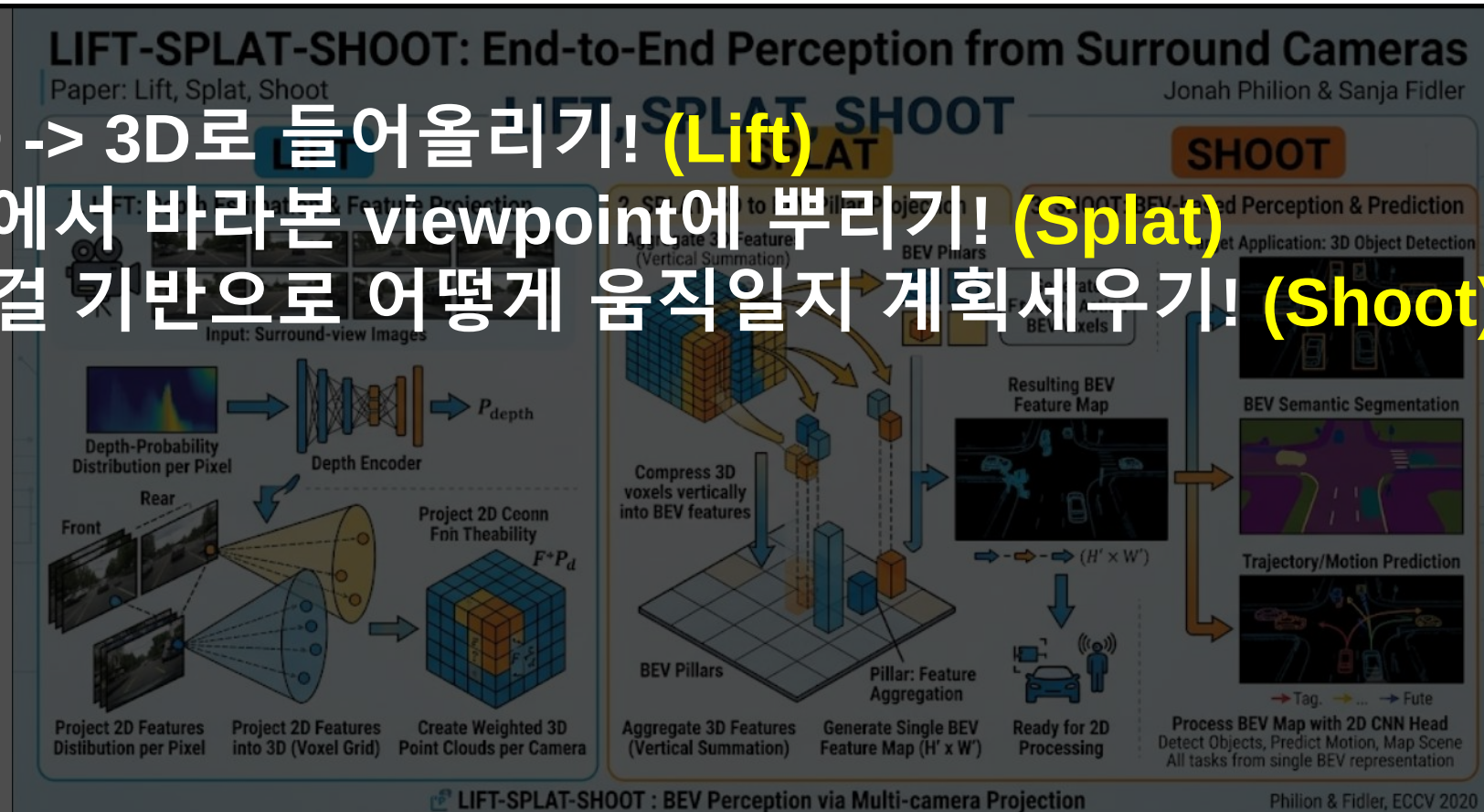
이론 설명



순방향투영방식: LSS(Lift-Splat-Shoot)

이론 설명

2D -> 3D로 들어올리기! (Lift)
위에서 바라본 viewpoint에 뿌리기! (Splat)
이걸 기반으로 어떻게 움직일지 계획세우기! (Shoot)



순방향투영방식: LSS(Lift-Splat-Shoot)

이론 설명

2D -> 3D로 들어올리기! (Lift)

위에서 바라본 viewpoint에 뿌리기! (Splat)

이걸 기반으로 어떻게 움직일지 계획세우기! (Shoot)

LIFT-SPLAT-SHOOT: End-to-End Perception from Surround Cameras

Paper: Lift, Splat, Shoot

Jonah Philion & Sanja Fidler

LIFT-SPLAT-SHOOT

Input: Surround-view Images

Depth-Probability Distribution per Pixel

Front Rear

Depth Encoder

Resulting BEV Feature Map

BEV Semantic Segmentation

Trajectory Prediction

Multi-camera Projection

Frustum of a Pyramid

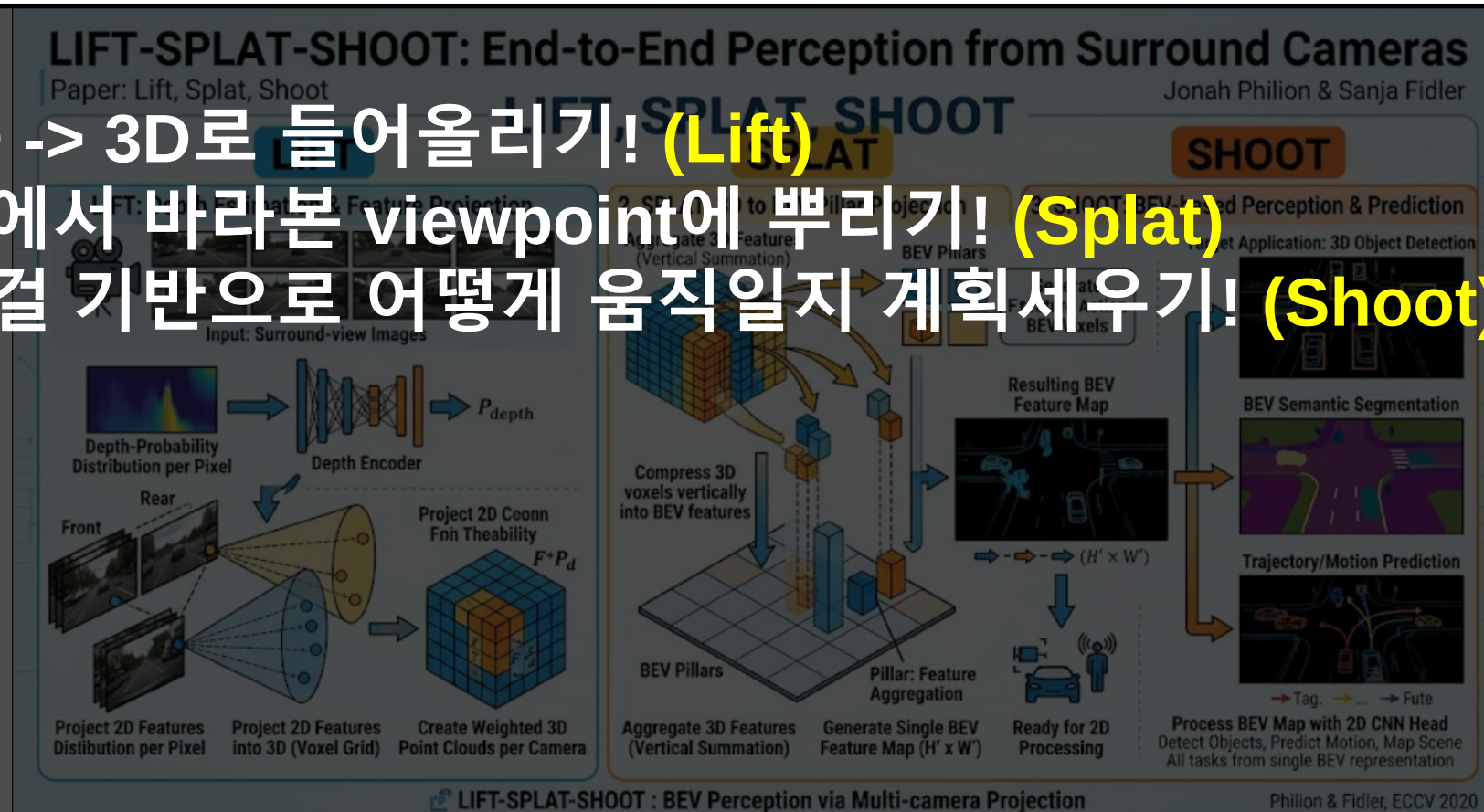
Depth Encoder

ACE Lab.

순방향투영방식: LSS(Lift-Splat-Shoot)

이론 설명

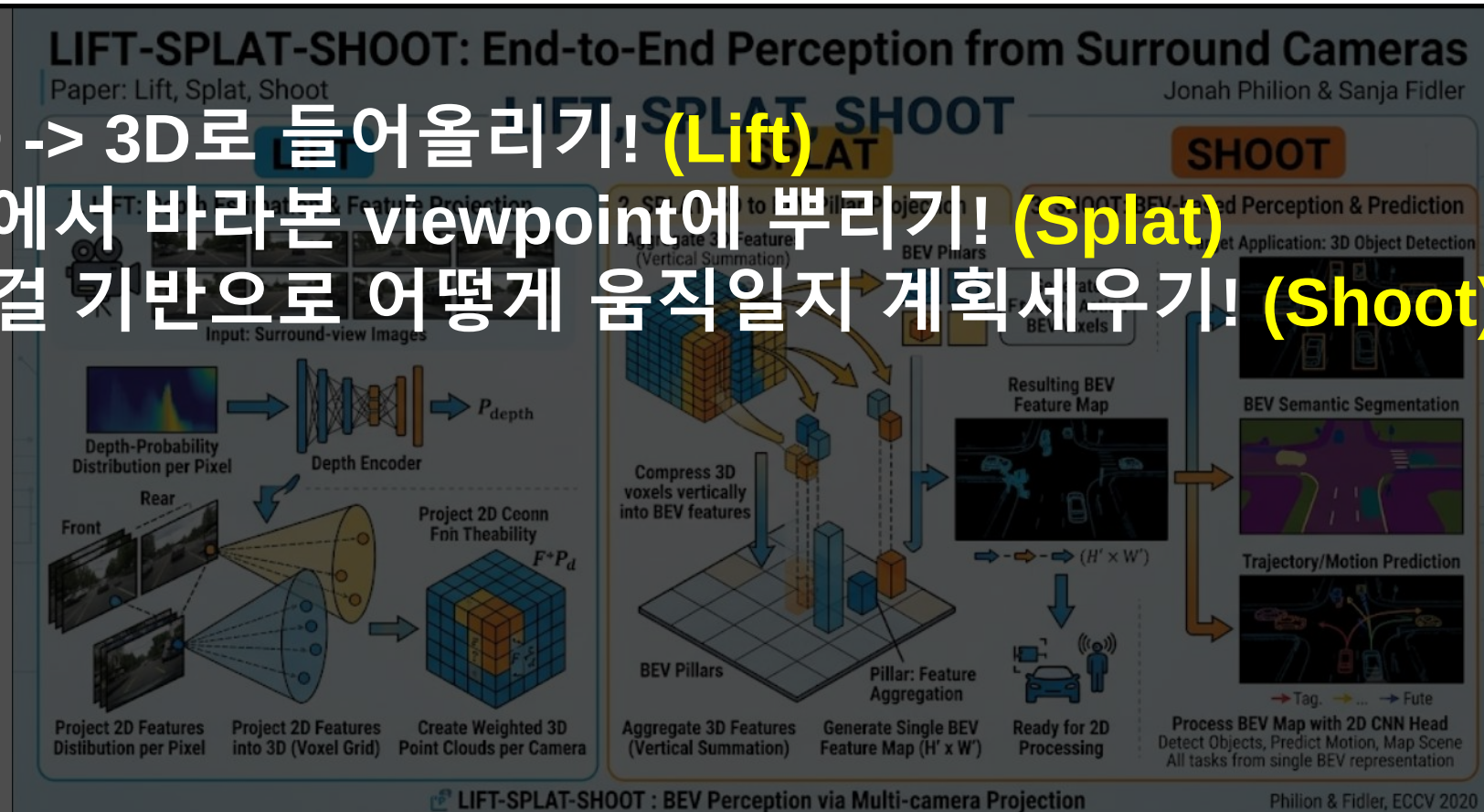
2D -> 3D로 들어올리기! (Lift)
위에서 바라본 viewpoint에 뿌리기! (Splat)
이걸 기반으로 어떻게 움직일지 계획세우기! (Shoot)



순방향투영방식: LSS(Lift-Splat-Shoot)

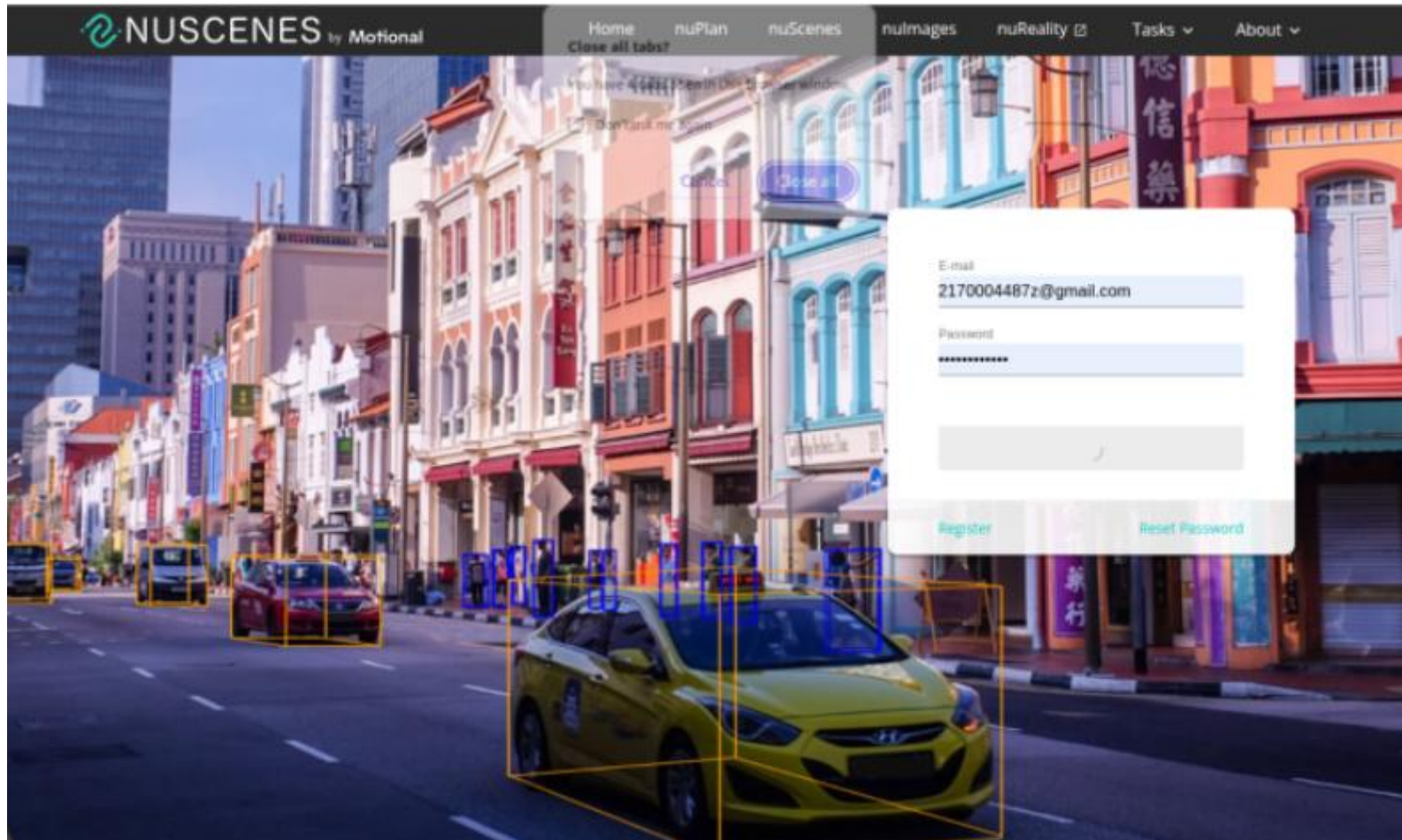
이론 설명

2D -> 3D로 들어올리기! (Lift)
위에서 바라본 viewpoint에 뿌리기! (Splat)
이걸 기반으로 어떻게 움직일지 계획세우기! (Shoot)



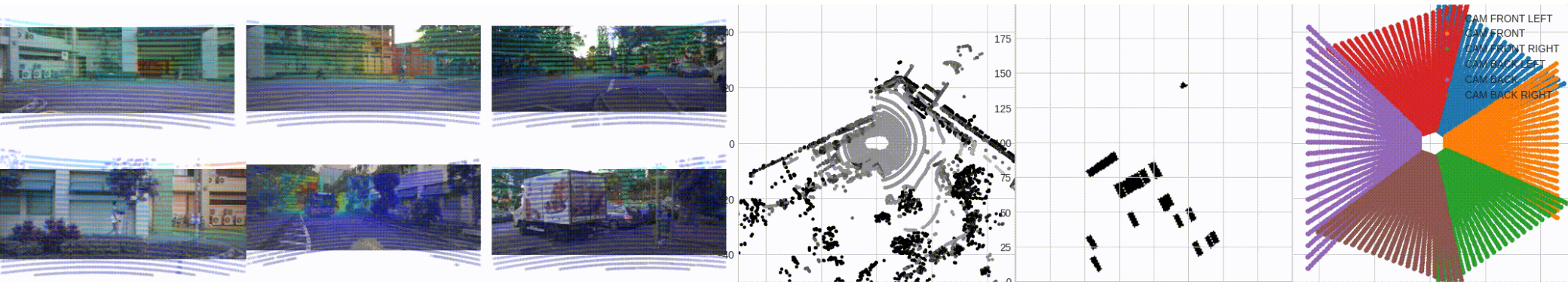
순방향투영방식: LSS(Lift-Splat-Shoot)

실습



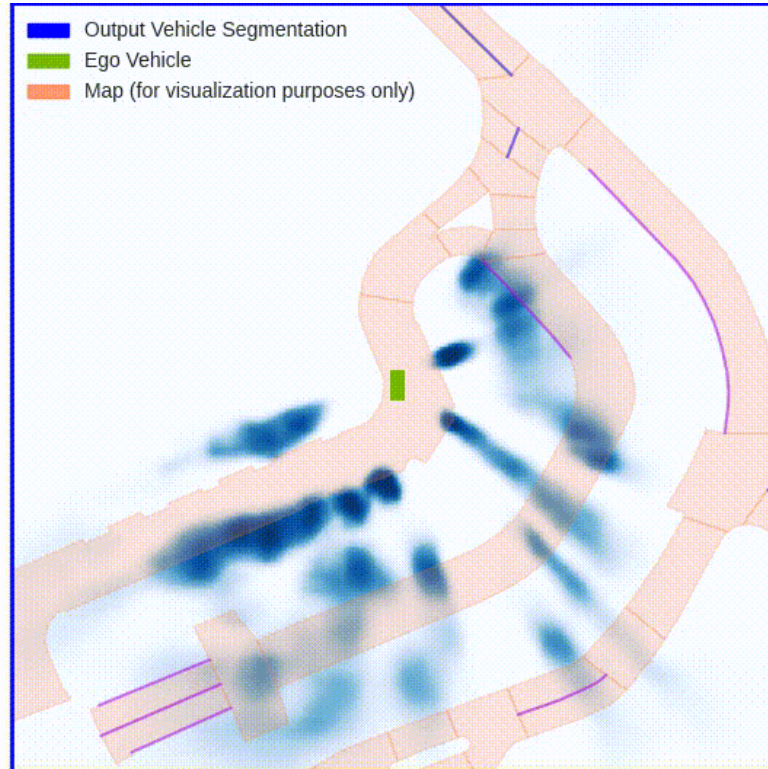
순방향투영방식: LSS(Lift-Splat-Shoot)

실습



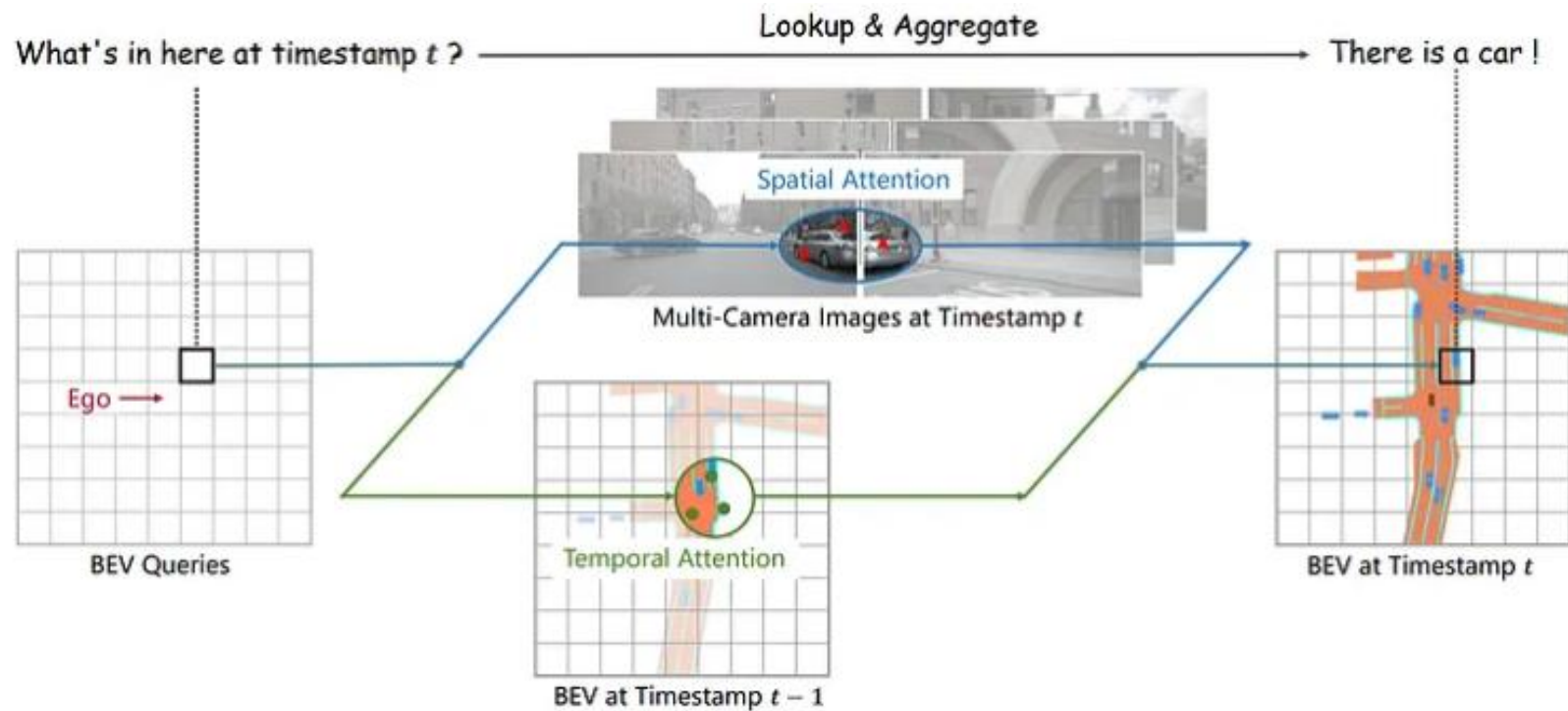
순방향투영방식: LSS(Lift-Splat-Shoot)

실습



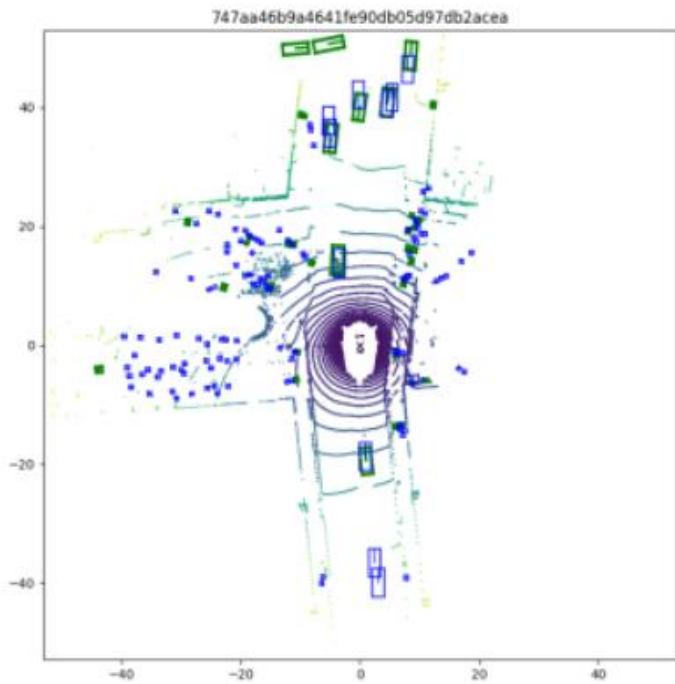
역방향투영방식: BEV Former

이론설명



역방향투영방식: BEV Former

실습



계획(Planning Stack)

※ 2가지 선택지 존재

1)경량 커스텀

7미션 상태머신 + Pure Pursuit 직접 구현. Nav2 오버헤드 회피

2)Nav2 채택

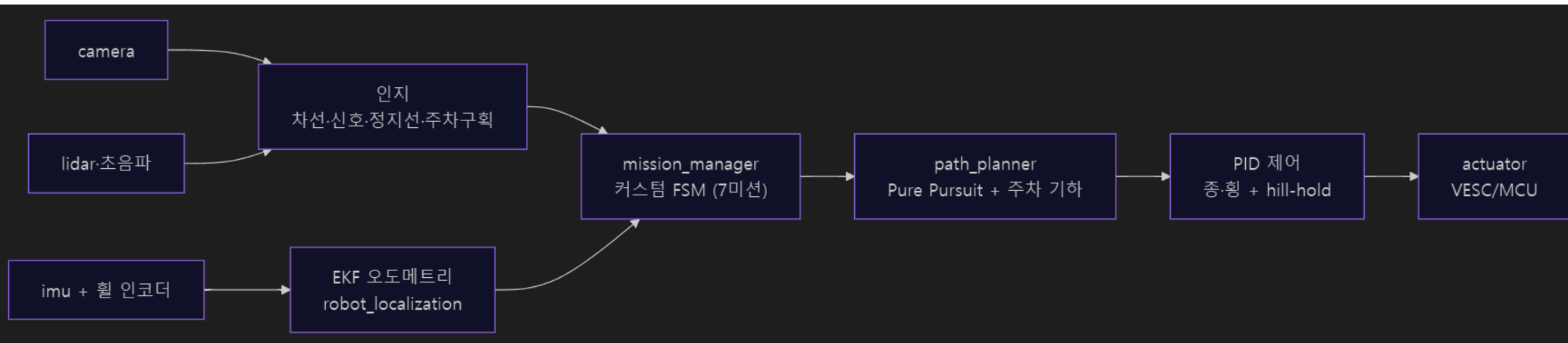
amcl · planner · controller · BT 활용. Ackerman 컨트롤러 설정 필요.

계획(Planning Stack)

※ 2가지 선택지 존재

1)경량 커스텀

7미션 상태머신 + Pure Pursuit 직접 구현. Nav2 오버헤드 회피



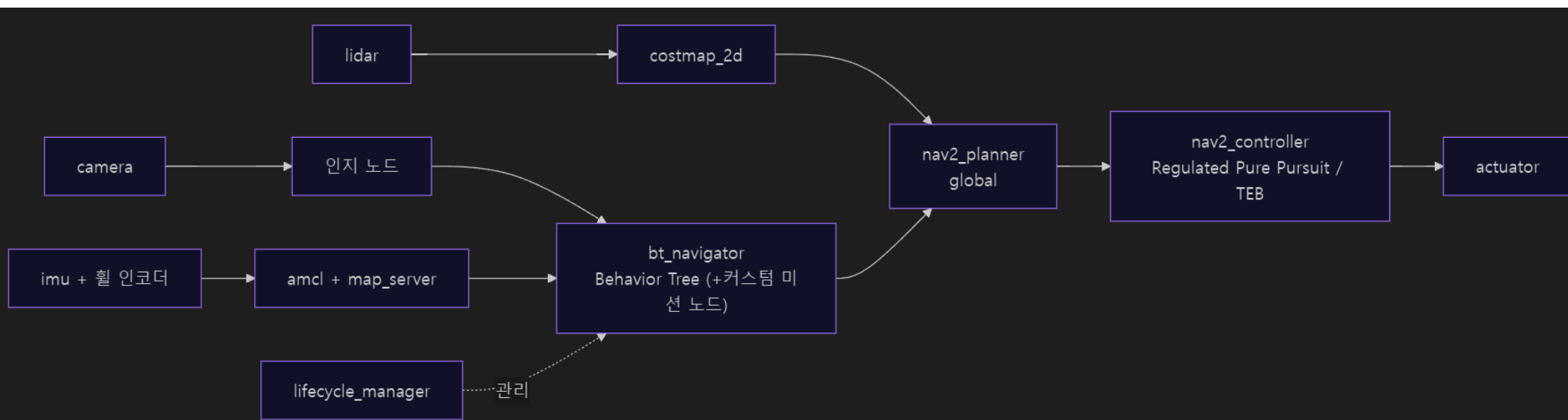
경량(Orin Nano 적합) · 코스 특화 최적화 · 디버깅/제어 흐름 단순 ·
의존성 최소 · 지연 예측 쉬움.

계획(Planning Stack)

※2가지 선택지 존재

2) Nav2 채택

amcl · planner · controller · BT 활용. Ackerman 컨트롤러 설정 필요.



검증된 스택

무거움(Orin Nano 부담) · yaml 설정 복잡 · 맵 필요(정형 코스엔 과할 수 있음) ·
Nav2는 goal-to-goal 내비 전제라 미션 시퀀스형 코스와 궁합 조정 필요

Q & A

Thank you for your attention



School of Electronics Engineering, KNU
ACE Lab.